MANSFIELD ISD

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PROFESSIONAL DEVELOPMENT END-OF-YEAR REPORT

FastAI Grader

End-of-Year Report

AI-Assisted Essay Grading

School Year 2025–2026

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Grade 6 ELA

April 2026

TEACHER

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SECTION 1

Executive Summary

At the start of the 2025–2026 school year, Shana Busby proposed a professional development goal centered on designing and piloting an AI-assisted reading and grading workflow for her Grade 6 ELA classes. The original vision described a broad platform encompassing book-based lesson planning, STAAR item import, Canvas/Google Docs publishing, adaptive learning, analytics dashboards, and AI-assisted grading — all delivered through a teacher-only tool with no student-facing interface.

Through iterative classroom piloting during the fall semester, Shana and her developer partner discovered that the **essay grading and high-quality student feedback** component delivered the most meaningful time savings and instructional value. Based on this real-world evidence, the project **pivoted** to focus entirely on building a production-grade AI essay grading platform — and that pivot paid off.

The result is a deployed, FERPA-compliant web application used daily in Shana's classroom that provides rubric-driven AI grading, inline text annotations, teacher-controlled feedback, and a secure student identity system — reducing grading turnaround from days to minutes while maintaining high-fidelity, evidence-based feedback for students.

PROFESSIONAL DEVELOPMENT GOALS — AT A GLANCE

Goal	Focus Area	Outcome
Goal 1	Curriculum Design Leadership	Met
Goal 2	Assessment Literacy	Met
Goal 3	Feedback & Grading Efficiency	Exceeded
Goal 4	Data-Informed Adjustments	Met

ARTIFACTS AVAILABLE FOR ADMINISTRATION REVIEW

1. **Application** — Live at ai-essaygrader.netlify.app (login credentials available from Shana)
2. **Graded Essay Samples** — Printed annotated PDFs with AI-generated feedback and teacher approvals
3. **Rubric Library** — Assignment rubrics created and refined throughout the year
4. **Dashboard Data** — Submission counts, grading statistics, and class period views
5. **CSV Exports** — Full submission data for gradebook integration
6. **This Report** — End-of-year summary with comparison to original goals

SECTION 2

Original Goals vs. Final Outcome

Status Key: Exceeded Met Adapted Not Built

Goal Area	Proposed	What Was Built	Status
AI-Assisted Essay Grading	AI suggests grades + comments tied to rubrics	Multi-provider LLM grading (Gemini, OpenAI, Anthropic) with BulletProof decimal-accurate scoring, rubric-driven feedback, teacher override	Exceeded
Evidence-Based Feedback	Feedback referencing text evidence, returned via Canvas	Inline annotations with line-by-line AI feedback, approve/edit/reject workflow, printable annotated PDFs	Exceeded
Rubric Management	Create/edit rubrics aligned to TEKS	Full rubric system: manual creation, AI enhancement, document upload & extraction (PDF/DOCX), rubric library per assignment	Exceeded
Teacher-in-Control	AI drafts; teacher reviews/approves	Complete teacher override on all grades and feedback, version history audit trail, teacher edits saved separately	Met
FERPA / Privacy	Student data stays in approved systems	Student Identity Bridge with AES-256-GCM encryption, zero PII in cloud, UUID-only storage, user-scoped data isolation	Exceeded
Multiple Input Methods	Import student work from Canvas/Docs	Direct upload: handwritten photos (AI Vision OCR via Gemini 2.5 Pro), DOCX files, plain text paste	Adapted
Dashboard & Data Views	Simple TEKS-by-skill summaries	Dashboard with search, filters, date ranges, class period organization, statistics cards, CSV export	Adapted
SSO / Authentication	Google or Microsoft SSO	Custom JWT authentication with email/password; multi-tenant support	Adapted
Book-Based Lesson Planning	RAG-powered lesson generation	Explored then intentionally abandoned — cost and complexity outweighed benefit	Not Built

Goal Area	Proposed	What Was Built	Status
STAAR Item Import	Parse STAAR items from URLs/PDFs	Not implemented — pivot to grading focus	Not Built
Canvas/Google Docs Integration	Publish assignments, sync grades	Not implemented — teachers use print/export instead	Not Built
Adaptive Learning	Personalized content by learning style	Not implemented — outside the grading focus	Not Built
TEKS Mastery Tracking	Aggregate performance by TEKS strand	Not implemented as standalone; rubric criteria can align to standards	Not Built

WHY WE PIVOTED

The original proposal envisioned a comprehensive platform. During the October–December pilot cycle, three key insights redirected development:

- **Grading was the bottleneck.** Shana spent the majority of her non-instructional time grading essays and writing feedback. Solving the grading problem had the highest return on time invested.
- **LMS integration added friction, not efficiency.** Canvas API integrations required district IT approval and significant development time for a workflow manageable via print/export.
- **Quality feedback was the student impact lever.** The inline annotation system produced better feedback than Shana could write manually in a fraction of the time.

The decision to pivot was a deliberate, **data-informed instructional adjustment** — exactly the kind of reflective practice outlined in the original professional development goals.

SECTION 3

What Was Built

CORE GRADING ENGINE

- **BulletProof Grading** — Deterministic decimal-precision scoring (decimal.js). LLM extracts rubric JSON; TypeScript computes with zero floating-point errors.
- **Multi-Provider LLM** — 16 models across OpenAI, Gemini, and Anthropic via Netlify AI Gateway.
- **Background Grading** — Trigger → background → polling eliminates timeouts on long essays.
- **Rubric-Driven** — Not ELAR-specific; a second teacher (AP World History) adopted it without modifications.

INLINE ANNOTATIONS & FEEDBACK

- **Two-Pass System** — Pass 1: general grading. Pass 2: criterion-specific inline feedback with line numbers, quotes, and suggestions.
- **Teacher Review Workflow** — Approve, edit, or reject each annotation. "Approve All" for bulk action.
- **Annotation Chat** — Teachers can ask follow-up questions about specific annotations.
- **Print Integration** — Annotated PDFs with highlights, corrections, strengths, and improvement areas.

STUDENT IDENTITY BRIDGE (FERPA)

- **Zero PII in Cloud** — Student names and district IDs are never transmitted to the server.
- **AES-256-GCM Encryption** — Student data encrypted locally with a teacher-chosen passphrase.
- **User-Scoped Isolation** — Each teacher's bridge is keyed to their user ID.
- **Class Period Organization** — Students organized by period with filtering throughout.
- **CSV Import** — Bulk student roster import.

MULTIPLE INPUT METHODS

- **Handwritten Essay Support** — AI Vision transcription using Gemini 2.5 Pro or GPT-4o.
- **DOCX Upload** — Client-side Word document parsing via mammoth.
- **Plain Text** — Direct paste for typed essays.
- **Local OCR Fallback** — Tesseract.js for offline handwriting recognition.

DASHBOARD & MANAGEMENT

- **Submission Dashboard** — Search, filter by assignment, class period, date range.
- **Statistics Cards** — Total assignments, submissions, pending review, graded today.
- **Three View Modes** — By Student, By Assignment, By Class.
- **CSV Export** — Full data export for gradebook integration.
- **Assignment Management** — Create, edit, delete assignments with rubric configuration.

AUTH, MULTI-TENANCY & QUALITY

- **Custom JWT Authentication** — Secure login with bcrypt password hashing.
- **Multi-Tenant Architecture** — Each teacher's data fully isolated by tenant ID.
- **582+ Automated Tests** — ~60% code coverage across unit and integration tests.
- **Version History** — Immutable audit trail for all grade changes.
- **Production Deployed** — ai-essaygrader.netlify.app with continuous deployment.

TECHNICAL ARCHITECTURE

Layer	Technology
Frontend	React + Vite, TypeScript, Tailwind CSS, shadcn/ui
Backend	Netlify Functions (serverless, Node.js)
Database	Neon Postgres (serverless, 7 tables, 21 indexes)
AI / LLM	Netlify AI Gateway — 16 models across OpenAI, Gemini, Anthropic
OCR	Gemini 2.5 Pro Vision (primary), Tesseract.js (fallback)
Privacy	Student Identity Bridge (AES-256-GCM, zero PII in cloud)
Testing	Vitest, 582+ tests, ~60% coverage
Hosting	Netlify (continuous deployment from GitHub)

SECTION 4

Outcomes & Impact

Goal 1 — Curriculum Design Leadership

Met

"Develop the ability to quickly produce and refine TEKS-aligned lesson materials tied to the actual texts used in class."

While the application did not include a lesson plan generator, Shana developed curriculum design skills through creating detailed rubrics for each assignment type — articulating clear, TEKS-aligned expectations for every writing assignment.

Evidence: A library of rubrics created in the application, each with criteria tied to Grade 6 ELA expectations.

Goal 2 — Assessment Literacy

Met

"Curate and adapt STAAR-formatted questions and rubrics; ensure items explicitly reference TEKS strands."

STAAR item import was not built, but assessment literacy grew significantly through creating and testing rubrics across 13 ELA document types, evaluating how each criterion aligned to TEKS strands.

Evidence: Rubric documents uploaded and extracted; side-by-side comparison of AI-scored vs. teacher-scored essays demonstrating rubric calibration.

Goal 3 — Feedback & Grading Efficiency

Exceeded

"Adopt an AI-assisted grading workflow (teacher-in-control) that returns consistent, evidence-based comments to students."

Fully achieved and exceeded. Grading turnaround reduced from 10–15 minutes manually to **2–3 minutes with AI assist + teacher review.** Inline annotations reference exact quotes with specific improvement suggestions.

Evidence: Graded essay samples with AI annotations and teacher approvals; annotated PDFs returned to students.

Goal 4 — Data-Informed Adjustments

Met

"Use artifacts and simple dashboards to monitor student progress and select targeted re-teaching or enrichment."

The Dashboard provides submission-level data with filtering by student, assignment, class period, and date range. CSV export enables integration with existing gradebook tools.

Evidence: Dashboard screenshots showing class-level views, statistics cards, and filtered submission lists.

STUDENT BENEFITS ACHIEVED

Proposed Benefit	Achieved?	Details
Clearer instructions and rubrics aligned to TEKS	Yes	Detailed rubrics with criteria, performance levels, and point values for every assignment
Faster, more consistent feedback pointing to specific passages	Yes	Inline annotations reference exact quotes with specific improvement suggestions
Differentiated materials via Canvas/Docs	No	Pivot to grading focus; differentiation was not the primary time-savings lever

Additional benefits: Higher-quality feedback (more detailed and consistent), faster turnaround enabling quicker revision cycles, and fair rubric-based scoring with mathematical precision.

TIMELINE RETROSPECTIVE

Period	Activity
Aug – Sep 2025	Set-up: Neon database, Netlify deployment, initial grading prototype. Baseline established for grading turnaround time.
Oct – Dec 2025	Pilot Cycle 1: Built BulletProof scoring, inline annotations, Student Bridge, Dashboard. Discovered grading was highest-value feature. Made pivot decision.
Jan – Mar 2026	Pilot Cycle 2: Refined rubric system, added AI Vision (Image-to-Text), class period organization. Second teacher (AP World History) adopted the tool.
Apr – May 2026	End-of-year report completed. Application stable and in daily use.

SECTION 5

Lessons Learned & Next Steps

LESSONS LEARNED

1. **Start with the pain point, not the feature list.** Classroom reality showed that one feature — grading with quality feedback — solved the biggest problem. Building that one thing well was more valuable than building everything partially.
2. **Pilot early, pivot quickly.** The Oct–Dec pilot cycle surfaced the pivot opportunity within weeks. By January, development was fully focused on the highest-impact feature set.
3. **Teacher-in-control is non-negotiable.** Every AI output in the system is a draft. The teacher reviews, approves, edits, or rejects. This principle was in the original proposal and remained central to the final product.
4. **FERPA compliance requires architectural commitment.** The Student Identity Bridge — encrypting all student data locally and sending only anonymous UUIDs to the cloud — ensures genuine FERPA compliance rather than policy-based compliance.
5. **A second user validates the design.** When a second teacher (AP World History) adopted the tool without modifications, it confirmed the rubric-driven architecture generalized beyond ELA — exceeding the original single-classroom scope.

RECOMMENDATIONS FOR 2026–2027

The FastAI Grader is stable, deployed, and in active use. Recommended next steps:

Priority	Item	Description
High	**Batch Upload**	Allow teachers to upload an entire class set of essays at once. Design complete — implementation pending.
High	**Expand Adoption**	Onboard additional ELA and cross-subject teachers, building on the AP World History success.
Medium	**Analytics**	Add per-rubric-criterion trend views to support data-informed re-teaching decisions.
Medium	**Revision Tracking**	Track improvement across multiple drafts of the same assignment to demonstrate student growth.
Pending	**Canvas Integration**	Revisit LMS integration if district IT approves API access — reduces manual export/upload steps.

Bottom line: A professional development goal that started as a broad AI platform proposal became a focused, impactful classroom tool — because teacher and developer stayed close to the real problem and were willing to change course when evidence demanded it.

Prepared by Shana Busby with development support from Mike Berry. · Live application: ai-essaygrader.netlify.app